

1164 Product Price at a Given Date

PLATFORM

Platform: LeetCode

DIFFICULTY

MEDIUM

DATE

Solved: July 1, 2026

Input

- 1 table — Products with columns product_id, new_price, change_date
- (product_id, change_date) is the primary key
- All products have default price **10** before any change

Output

- Every product_id with its price **on 2019-08-16**
- If changed on or before date → most recent price
- If never changed before date → default price = **10**

Key Insights

- **2 cases** exist → use UNION to combine them
- Case 1 → find most recent price change on or before target date using MAX(change_date)
- Case 2 → products with no changes before target date → default price = 10
- Pair matching (product_id, change_date) IN (subquery) gets the correct price row

Observations

- A product can have multiple price changes → only the **most recent** one counts
- Future price changes (after 2019-08-16) must be **ignored** → WHERE change_date <= date
- Products with only future changes → treated as if never changed → price = 10
- NOT IN safely identifies products with no valid price history

Approach

2 cases handled separately then combined:

Case 1 → Products WITH changes on or before date:

- Find MAX(change_date) per product
- Match back to get new_price

Case 2 → Products WITHOUT changes on or before date:

- Use NOT IN to find them
- Assign price = 10

Combined → UNION

Algorithm

Query 1:

Inner query: GROUP BY product_id

MAX(change_date) <= '2019-08-16'

Outer query: WHERE (product_id, change_date) IN (inner)

→ gets most recent price per product

UNION

Query 2:

WHERE product_id NOT IN (

products that have change_date <= '2019-08-16'

)

→ gives products with default price 10

Final Result → all products covered!

Time & Space Complexity

	Complexity
TC	O(N log N) — grouping and sorting by date
SC	O(N) — storing intermediate results

Database

Description

Editorial

Solutions

Submis

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SQL SchemaPandas Schema

Table: Products

Column Name	Type
product_id	int
new_price	int
change_date	date

(product_id, change_date) is the primary key (combination of columns with unique values) of this table.
Each row of this table indicates that the price of some product was changed to a new price at some date.

Initially, all products have price 10.

Write a solution to find the prices of all products on the date 2019-08-16.

Return the result table in **any order**.

The result format is in the following example.

Example 1:

1.4K22932 Online

Accepted

All Submissions

Accepted 15 / 15 testcases passed

Sahil Khan submitted at Jul 01, 2026 19:35

Runtime

480 msBeats 68.43%

Code | MySQL

```
1 # Write your MySQL query statement below
2 SELECT product_id, new_price AS price
3 FROM Products
4 WHERE (product_id, change_date) IN (
5     SELECT product_id, MAX(change_date)
6     FROM Products
7     WHERE change_date <= '2019-08-16'
8     GROUP BY product_id
9 )
10
11 UNION
12
13 SELECT product_id, 10 AS price
14 FROM Products
15 WHERE product_id NOT IN (
16     SELECT product_id
17     FROM Products
18     WHERE change_date <= '2019-08-16'
19 )
```

Accepted Runtime: 84 ms

Case 1

Input

Products =